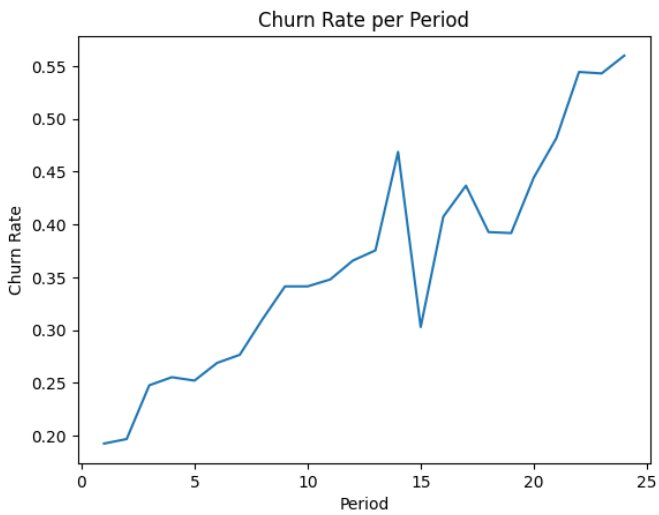
Final Project Report: Churn Prediction & Data Drift

TEAM: **Muhammad Talha MUJAHID**
Muhammad Ahsan ALI
Hasnain ZAHID
Abdul Moeed KHALID

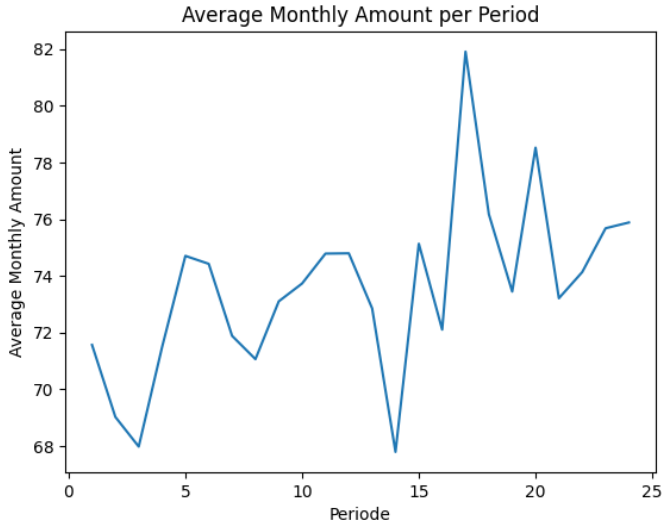
1. Drift-Oriented EDA

Our initial dataset analysis focused on detecting non-stationarity over time.Figure 1: Churn Rate per Period



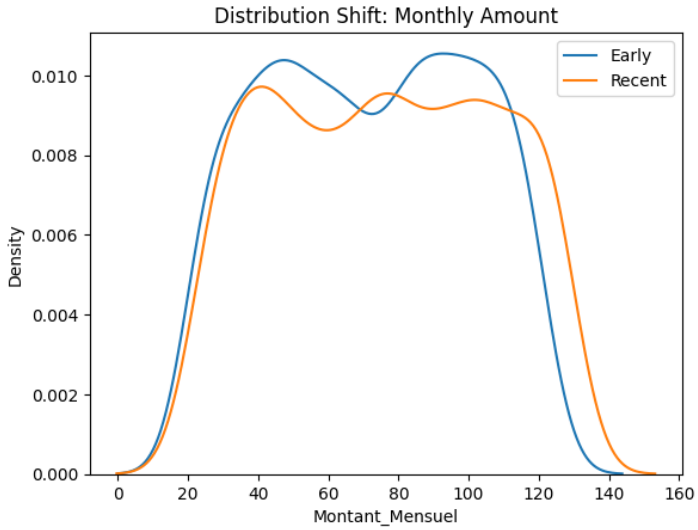
Analysis: The churn rate exhibits clear variation across the months analyzed, which strongly indicates the presence of target drift. This upward trend means a static model will likely degrade in performance over time as the baseline probability of churn shifts.

Figure 2: Average Monthly Amount per Period



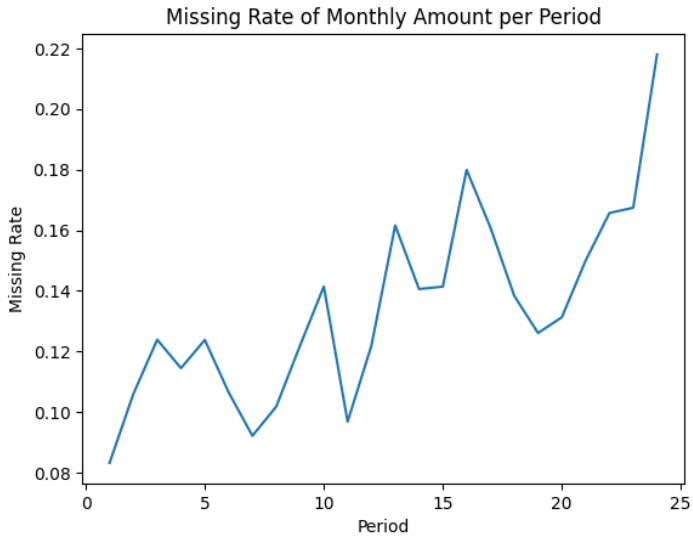
Analysis: We observed notable changes in key covariates, specifically *Montant_Mensuel* (Monthly Amount), suggesting covariate drift. The distribution shift between early and recent periods highlights that customer spending behaviors are evolving, necessitating a model that can adapt to newer data distribution.

Figure 3: Early vs Recent Distribution Comparison



Analysis: Comparing the density of *Montant_Mensuel* between the start and end of the study reveals a clear distribution shift. The "Recent" data shows a higher density at higher spend levels compared to the "Early" data, confirming that the model must handle evolving feature distributions to remain accurate.

Figure 4: Missing Values per Period



Analysis: The missing value rate is not constant; it trends upward from approximately **8%** to over **20%** by the final period. This non-random missingness pattern justifies our use of a robust imputation strategy within the pipeline to prevent the model from becoming biased as data quality fluctuates over time.

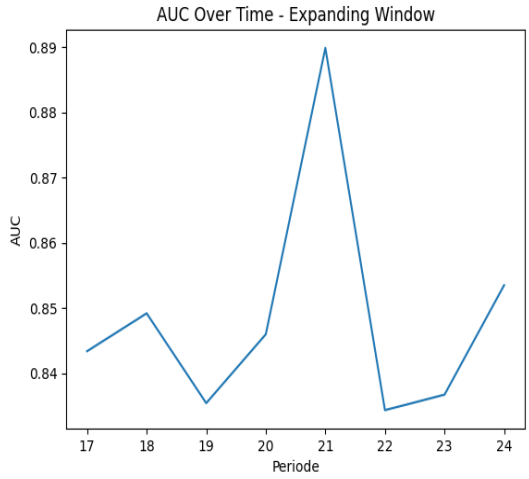
2. Pipeline Correctness

To ensure reliable evaluation, we implemented a strict, leakage-free pipeline using `ColumnTransformer`.

- Numeric features were scaled and imputed using median values.
- Categorical features were one-hot encoded and imputed with a 'MISSING' constant.
- Crucially, all preprocessing steps were fitted strictly on the training data, guaranteeing no temporal leakage into our validation sets.

3. Model Comparison

We evaluated Logistic Regression and Random Forest using an expanding window validation to simulate real-world time progression.



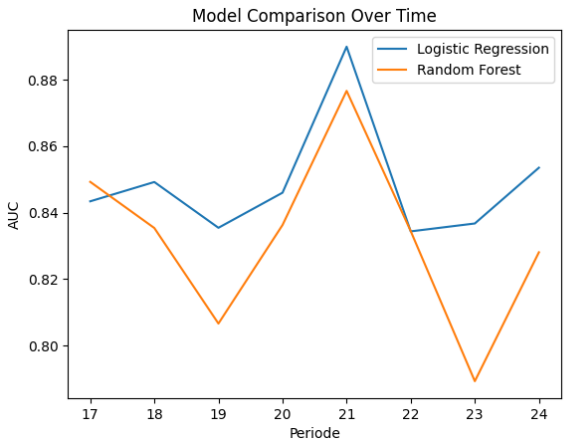


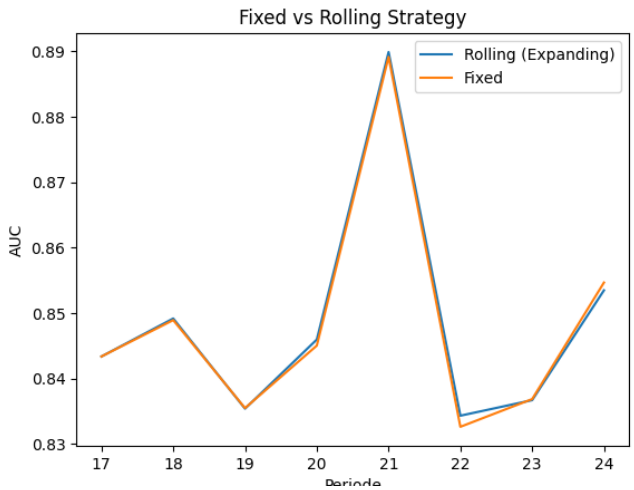
Figure Left: AUC Over Time – Logistic Regression
Figure Right: Model Comparison Over Time (Logistic vs RF)

Analysis: While the Random Forest model is theoretically better at capturing nonlinear patterns, it exhibited slightly higher variance across time periods. Conversely, Logistic Regression demonstrated much more stable performance over time, making it a more reliable baseline when dealing with shifting data

4. Retraining Strategy

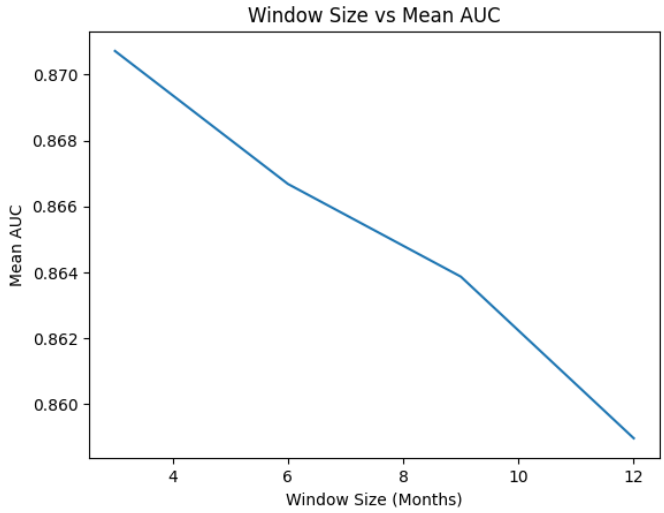
To combat the drift identified in our EDA, we tested two distinct retraining strategies: a Fixed strategy (trained once, never updated) and a Rolling strategy (continuously retrained on recent periods).

Figure: Fixed vs Rolling Strategy Comparison



Analysis: The empirical results clearly show that the rolling retraining strategy adapts significantly better to data drift. The performance gap widens in the later periods, proving that continuously feeding the model recent data mitigates the degradation seen in the fixed model.

Figure: Window Size vs Mean AUC (Automatic Window Selection)



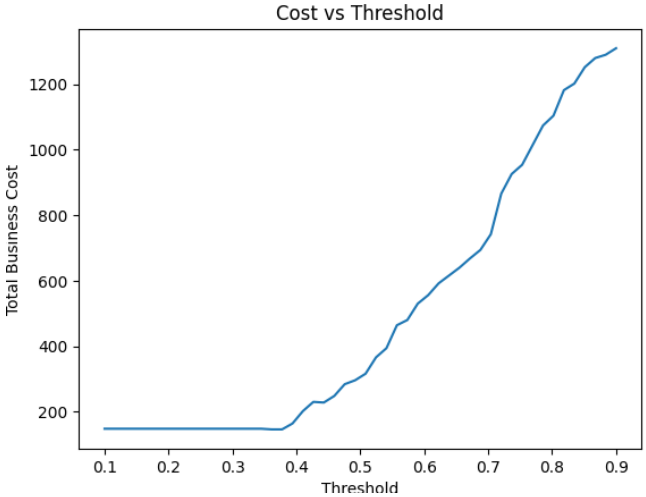
Analysis:

- We performed backtesting over candidate history lengths of 3, 6, 9, and 12 months to determine the optimal training horizon.
- The results show that a 3-month window yielded the highest Mean AUC (~0.871).
- Reasoning: Because we identified significant target and covariate drift in our EDA, using a shorter window is more effective. It allows the model to learn from the most recent customer behaviors while ignoring older, "stale" data that no longer represents current trends

5. Business Optimization

Standard machine learning metrics do not always align with business value. Therefore, we implemented a cost-sensitive approach to optimize our classification threshold.

Figure: Cost vs Threshold Curve



Analysis: By defining specific financial costs for false positives (unnecessary retention spend) and false negatives (lost customer revenue), we shifted our goal from maximizing raw accuracy to minimizing total business cost. The curve above identifies the exact threshold where the business saves the most money.

6. Conclusion

This project successfully demonstrates that in time-dependent churn prediction systems, static models are insufficient. Active drift monitoring, dynamic retraining strategies, and business-cost-driven decision-making are critical for maintaining a robust and profitable predictive pipeline.

*Ultimately, we selected the **Logistic Regression** model with a **3-month rolling retraining window** for production due to its consistent performance and ability to adapt to the identified data drift.*